Tutorial-3. 10/02/2026.

(Linear Regression).

Car age(x_1)	Mileage(x_2)	Resale Price(y)
2	20	6.
3	35	7.

$\Rightarrow y = mx + c.$

$y = c + m_1 x_1 + m_2 x_2.$

$\Downarrow$

$6 = c + m_1 \cdot 2 + m_2 \cdot 3$

$7 = c + m_1 \cdot 3 + m_2 \cdot 4.$

Matrix form:

$$\underset{y \; 2\times1}{\begin{pmatrix} 6 \\ 7 \end{pmatrix}} = \underset{X \; 2\times3}{\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \end{pmatrix}} \underset{\beta \; 3\times1}{\begin{pmatrix} c \\ m_1 \\ m_2 \end{pmatrix}}$$

$$\boxed{y = X\beta}$$

$\Rightarrow$ multiply X^T to the both sides.

$X^T y = (X^T X)\beta.$

$(X^T X)^{-1} = (X^T y) = \beta$ $\qquad \left[(X^T X)^{-1}(X^T X) = 1\right]$

$$X^T = \begin{pmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{pmatrix}$$

$$X^T \cdot X = \begin{pmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 1+1 & 2+3 & 3+4 \\ 2+3 & 4+9 & 6+12 \\ 3+5 & 6+15 & 9+16 \end{pmatrix} = \begin{pmatrix} 2 & 5 & 7 \\ 5 & 13 & 18 \\ 8 & 21 & 25 \end{pmatrix}$$

$$X^T.X = \begin{pmatrix} 2 & 5 & 7 \\ 5 & 13 & 18 \\ 8 & 21 & 25 \end{pmatrix} = \begin{vmatrix} 2 & 5 & 10 \\ 5 & 13 & 21 \\ 8 & 21 & 34 \end{vmatrix}$$

$$(X^T X)^{T} = \frac{\operatorname{adj}(X^T X)}{\det(X^T X)}.$$

$$\det(X^T X) = 2(13\times 34 - 21\times 21) - 5(5\times 34 - 8\times 21) + 10(5\times 21 - 8\times 13)$$

$$= 2 - 10 + 10.$$

$$\det(X^T X) = 2.$$

$$\operatorname{adj}(X^T X) = \begin{bmatrix} +\begin{vmatrix} 13 & 21 \\ 21 & 34 \end{vmatrix} & -\begin{vmatrix} 5 & 21 \\ 8 & 34 \end{vmatrix} & +\begin{vmatrix} 5 & 13 \\ 8 & 21 \end{vmatrix} \\ -\begin{vmatrix} 5 & 10 \\ 21 & 34 \end{vmatrix} & +\begin{vmatrix} 2 & 10 \\ 8 & 34 \end{vmatrix} & -\begin{vmatrix} 2 & 5 \\ 8 & 21 \end{vmatrix} \\ +\begin{vmatrix} 5 & 10 \\ 13 & 21 \end{vmatrix} & -\begin{vmatrix} 2 & 10 \\ 5 & 21 \end{vmatrix} & +\begin{vmatrix} 2 & 5 \\ 5 & 13 \end{vmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} (13\times 34 - 21\times 21) & (5\times 34 - 8\times 21) & (5\times 21 - 13\times 8) \\ (5\times 34 - 21\times 10) & (2\times 34 - 10\times 8) & (2\times 21 - 8\times 5) \\ (5\times 21 - 10\times 13) & (2\times 21 - 5\times 10) & (2\times 13 - 5\times 5) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & 1 \\ -40 & -12 & 2 \\ -25 & -8 & 1 \end{bmatrix}$$

$$(x^T x)^{-1} = \frac{\text{adj}\,(x^T x)}{\det\,(x^T x)}.$$

$$x^T y = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 5 \end{bmatrix}_{3\times 2} \begin{bmatrix} 6 \\ 7 \end{bmatrix}_{2\times 1} = \begin{bmatrix} 6+7 \\ 12+21 \\ 18+35 \end{bmatrix} = \begin{bmatrix} 13 \\ 33 \\ 53 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & 20 & -12.5 \\ -1 & -6 & 4 \\ 0.5 & -1 & 0.5 \end{bmatrix}_{3\times 3} \begin{bmatrix} 13 \\ 33 \\ 53 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5\times 13 + 20\times 33 + (-12.5\times 53) \\ -1\times 13 + -6\times 33 + 4\times 53. \\ 0.5\times 13 + -1\times 33 + 0.5\times 53 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \beta. \qquad \begin{aligned} c &= 4 \\ m_1 &= 1 \\ m_2 &= 0 \end{aligned}$$

$$y^n = c + m_1 x_1 + m_2 x_2.$$

$$\boxed{y^n = 4 + 1\cdot x_1 + 0 x_2}$$

Pred.

$$y_1^n = 4 + 2\cdot(1) + 3(0). \qquad \boxed{y^n_{Pred} = 6.}$$

$\hat{y}_2 = 4 + (3)1 + 0.(5)$ } . as these values are equal to actual values so then α is "0".

$= 7$.

$$\boxed{\hat{y}_2 = 7 \\ \text{Pre}}$$

Now calculate the α value.

→ Root mean square Error:

$\Rightarrow \frac{1}{N} \sum_{i} (y_i - \hat{y}_i)^2$

$= \frac{1}{2} \sum_{i} ((0)^2 + (0)^2)$

$= \frac{1}{2} \left[(0)^2 + (0)^2 \right]$

$= 0$

Ex:- $\hat{y} = 2$, $\hat{y} = 7$ then

$= \frac{1}{2}\left[(6-2)^2 + (7-7)^2\right]$

$= \frac{1}{2}\left[(4)^2 + (0)^2\right]$

$= \frac{1}{\cancel{2}} \cancel{16}\ 8$

$= 8$.

① Which variable act as Predictors?

$\Rightarrow x_1$ is important as x_2 coefficient is '0' so do not importance of x_2.